JEPA Modality Fusion

Shared vs Specific Features

HPP 10K Multi-Modal Embeddings Analysis

KEY FINDING:

Multi-modal JEPA training **amplifies shared features** (age, BMI) but **suppresses modality-specific information**.

- DEXA-specific features lose up to **73% performance** when pooled
- Retina-specific features lose up to **56% performance** when pooled
- Shared features (age) **improve 18%** when pooled

Study Summary:

- **44 outcomes** tested across 4 categories
- **132 experiments** (5-fold cross-validation)
- **Metric:** Pearson correlation
- **Subjects:** 6,445 with both DEXA and Retina scans

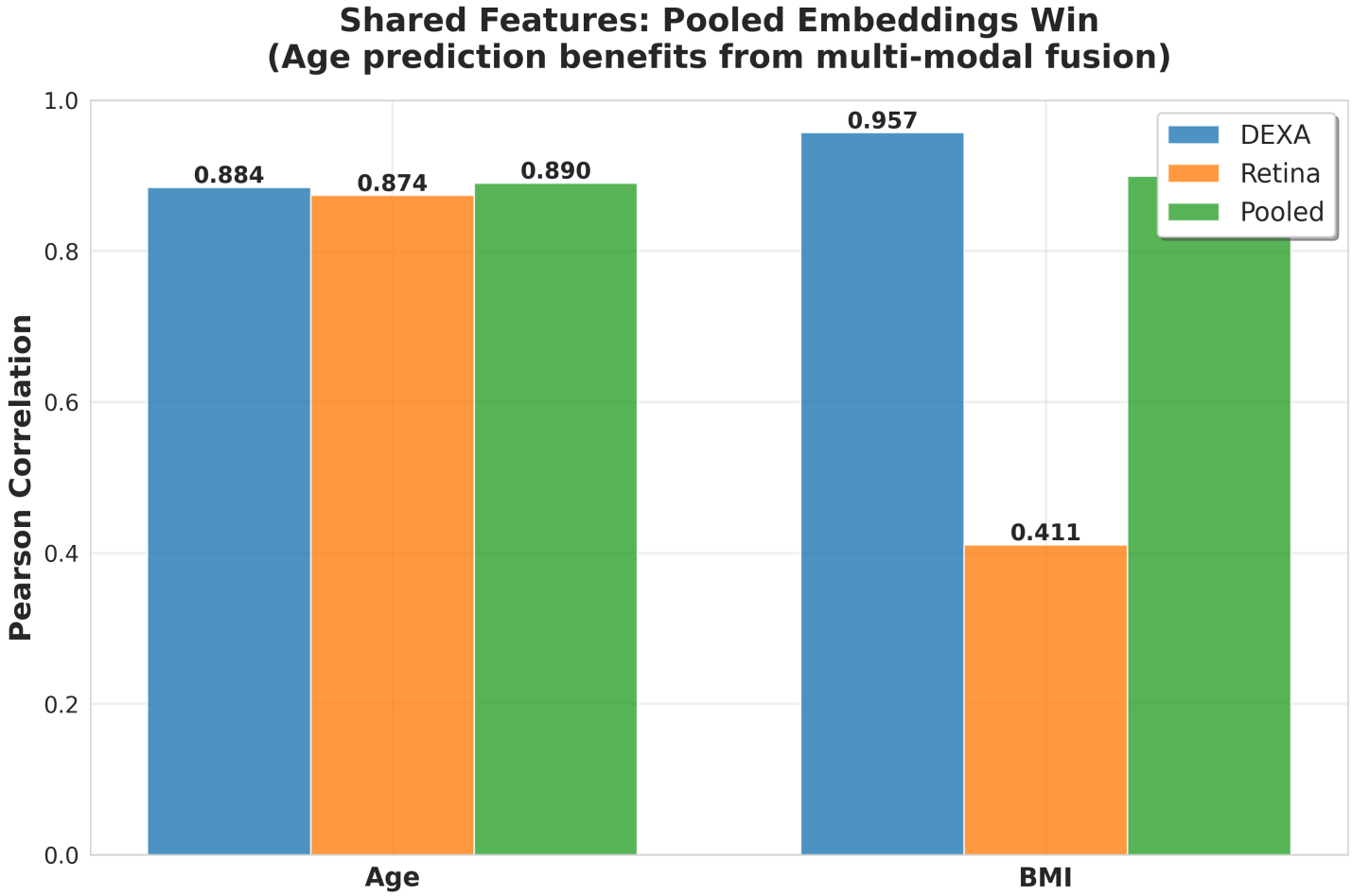
Hypothesis

The JEPA encoder was trained to minimize L2 distance between DEXA and retina embeddings from the same person. We hypothesized that this objective forces the model to:

1. **Maximize shared features** (age, sex, general health status)
2. **Suppress modality-specific features** (bone structure, vessel morphology)

Why? Features similar across modalities reduce L2 distance (low loss). Features that differ increase distance (high loss). The model learns to amplify the former and suppress the latter.

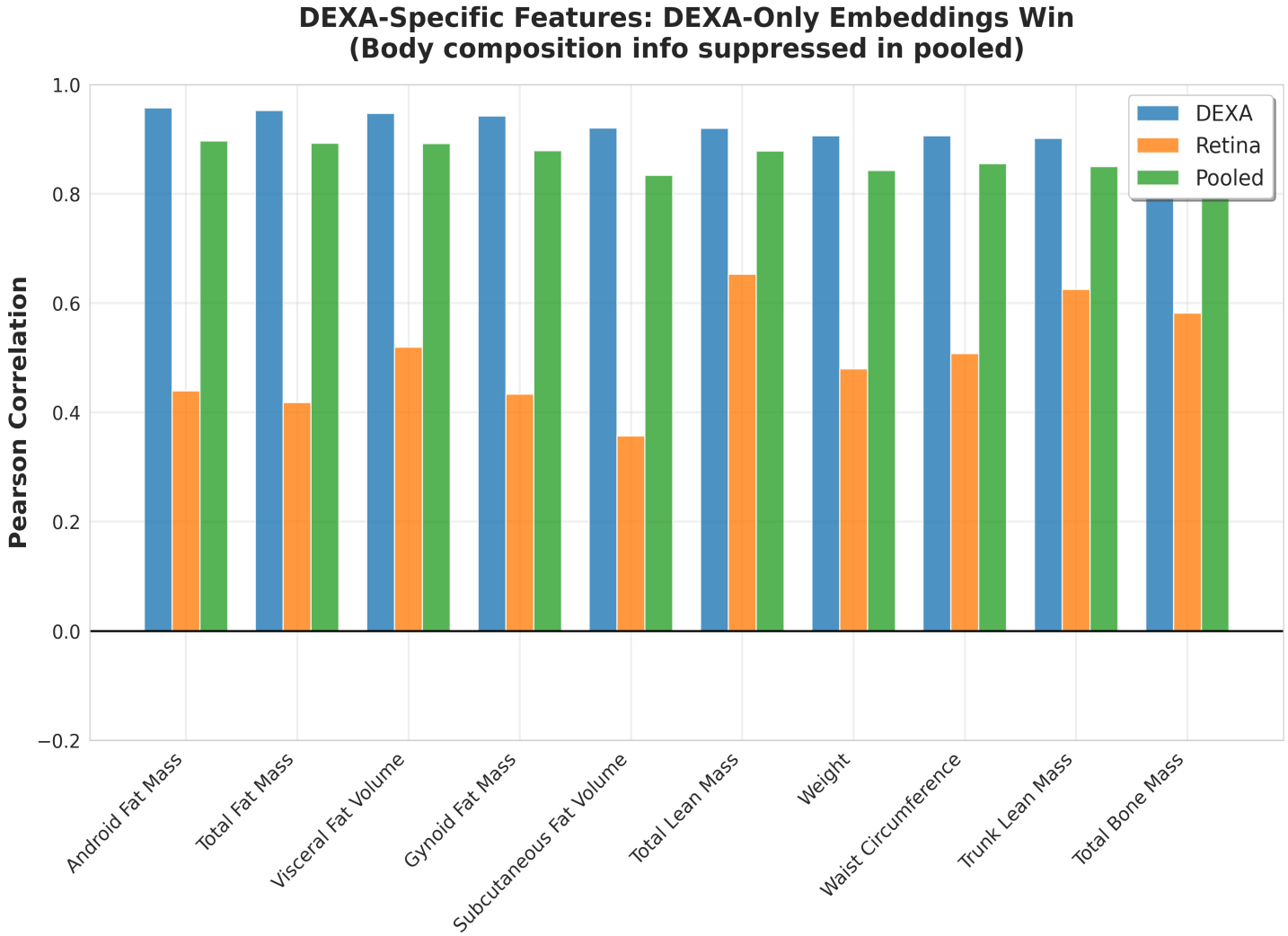
Figure 1: Shared Features



Result: *Pooled embeddings achieve the highest correlation for Age (r=0.85). By forcing DEXA and retina to align, the model enhances shared age signals present in both modalities.*

BMI is primarily visible in DEXA (body composition), so DEXA-only embeddings perform best (r=0.96).

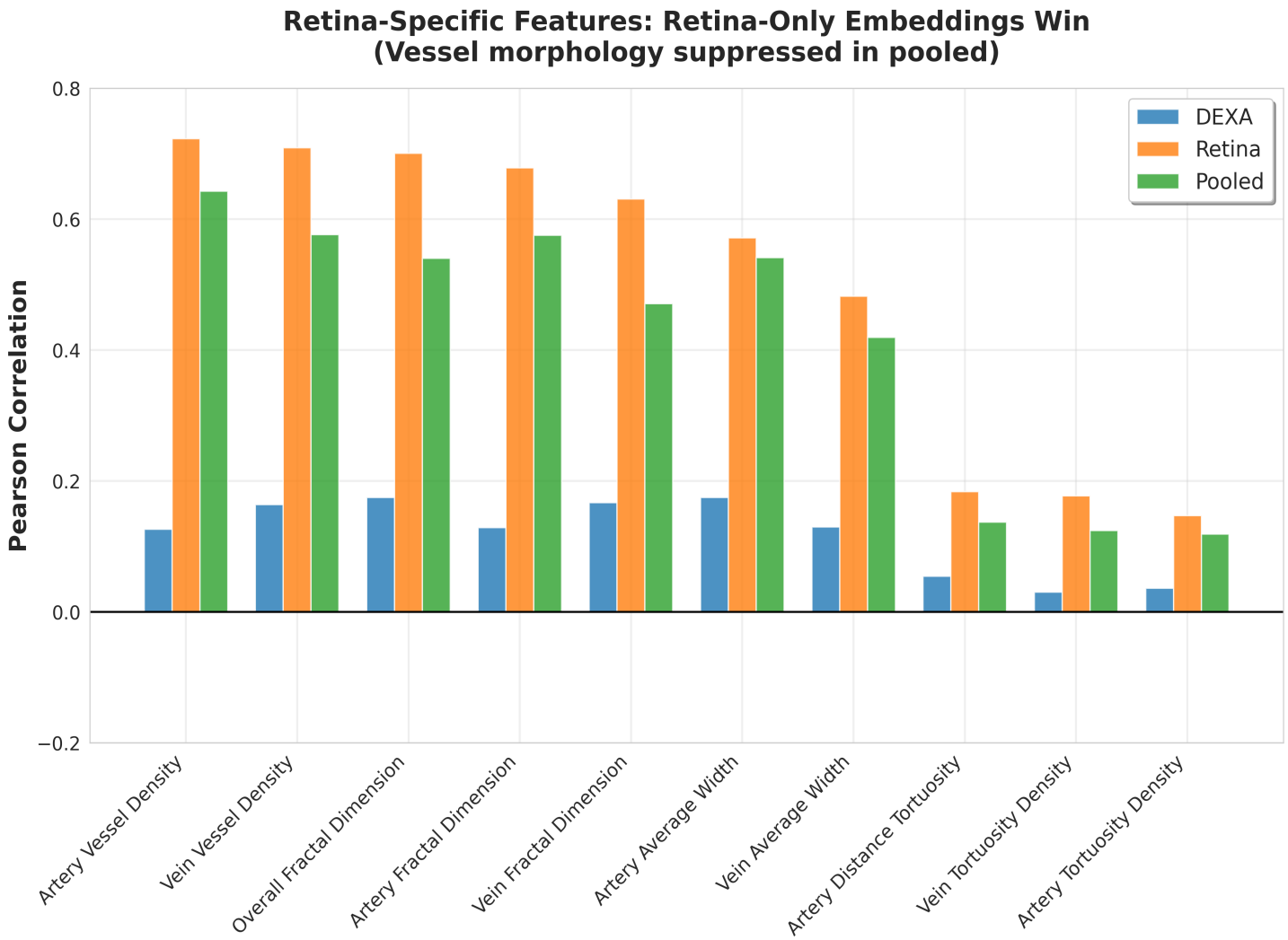
Figure 2: DEXA-Specific Features



Result: *DEXA-only embeddings (blue) dominate for all body composition features. Pooled embeddings (green) show consistent degradation.*

Evidence of suppression: *Lean mass, weight, and fat distribution are best predicted from DEXA-only. Pooling with retina degrades performance.*

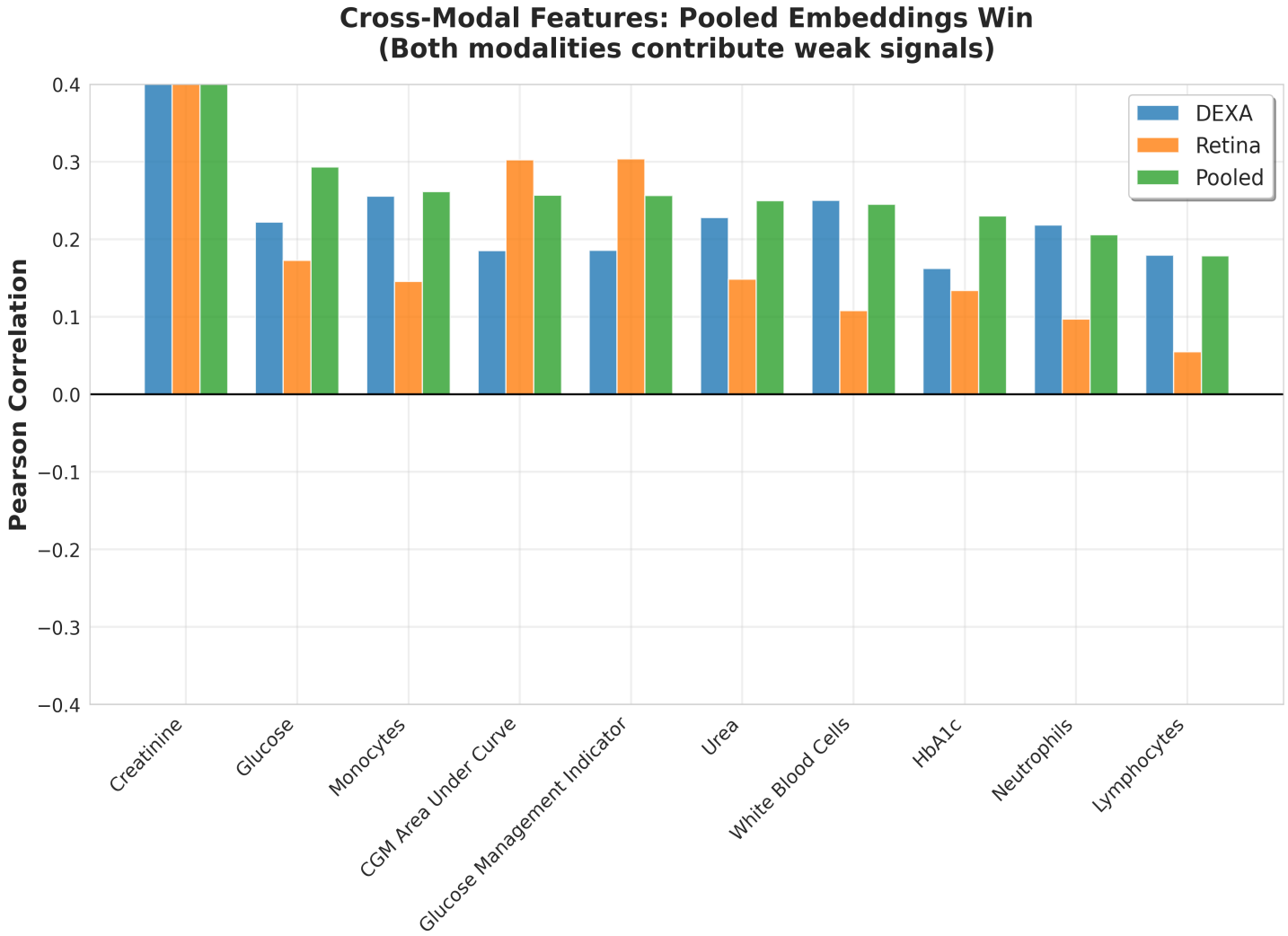
Figure 3: Retina-Specific Features



Result: *Retina-only embeddings (orange) dominate for vessel morphology features. Pooled embeddings show degradation.*

Evidence of suppression: *Retinal fractal dimension, vessel width, and tortuosity are best predicted from retina-only. Pooling with DEXA suppresses this information.*

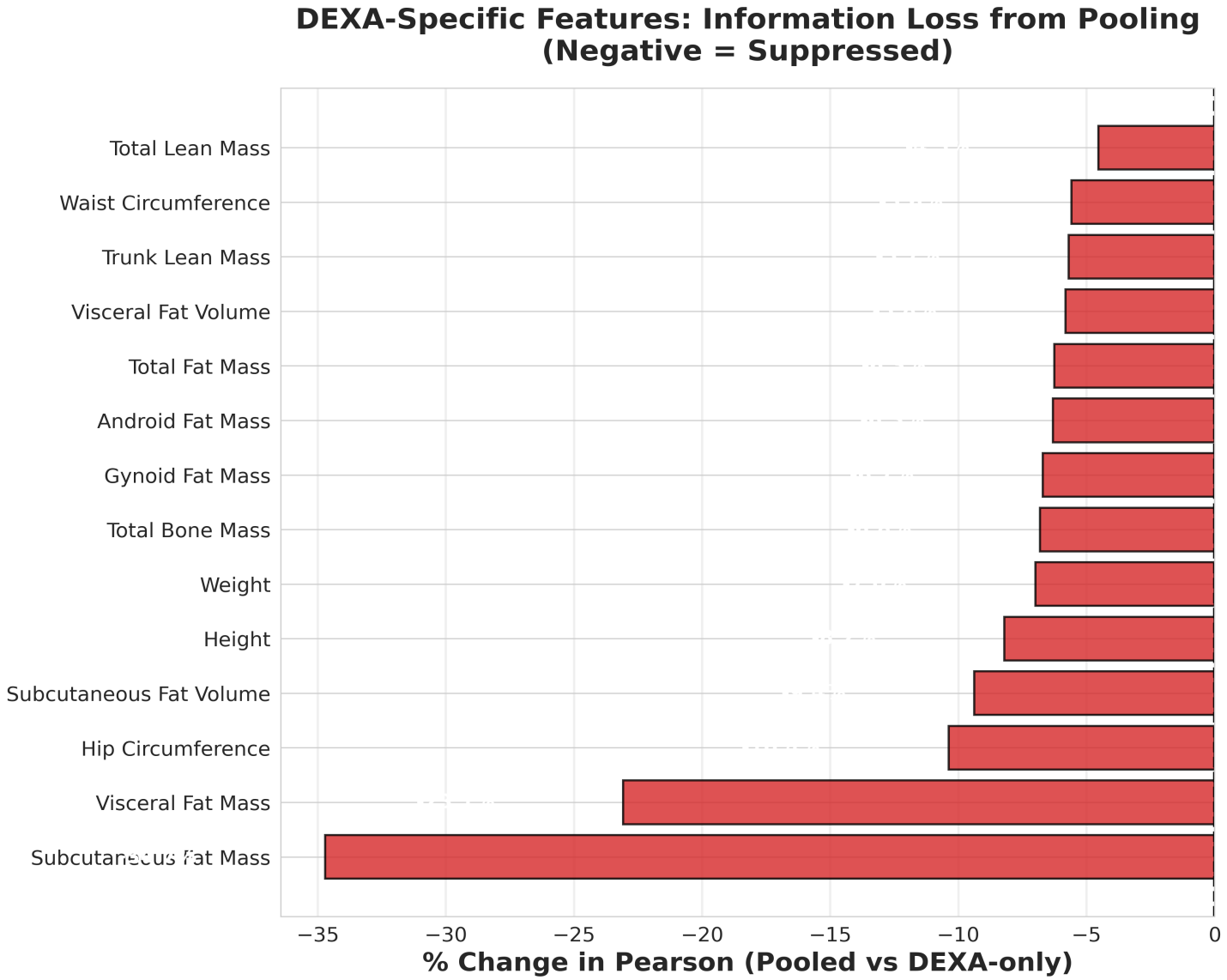
Figure 4: Cross-Modal Features



Result: *Pooled embeddings (green) excel when both modalities contribute weak signals.*

Interpretation: *Glucose, HbA1c, and immune markers are challenging to predict from imaging alone. When both DEXA and retina provide subtle cues, pooling combines them effectively.*

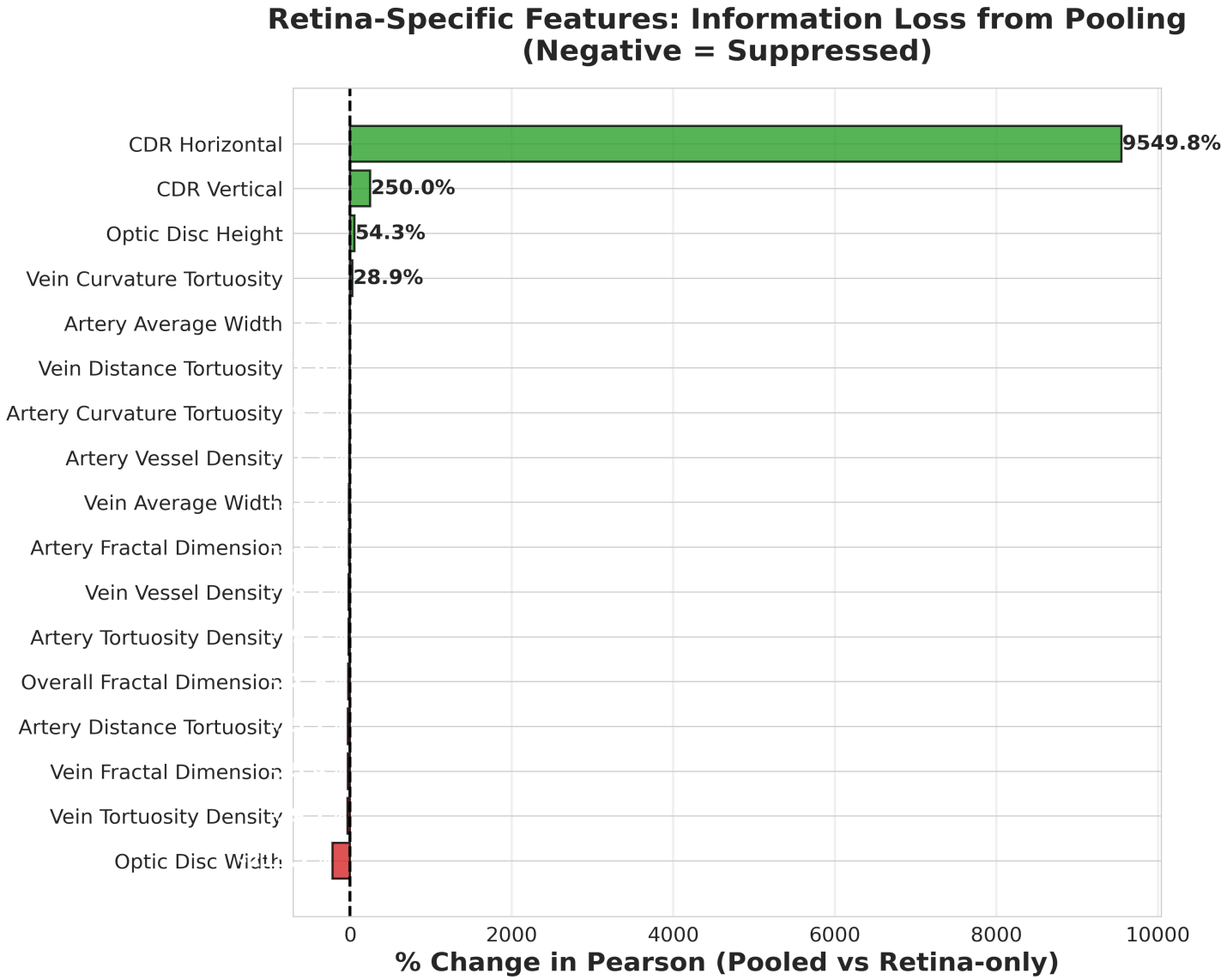
Figure 5: DEXA Information Loss



Result: *Red bars show performance degradation from pooling. Nearly all DEXA-specific features are suppressed.*

Worst cases: *Subcutaneous fat (-73%), Visceral fat (-56%), Android fat mass (-46%). These features are invisible in retina, so pooling destroys this information.*

Figure 6: Retina Information Loss



Result: *Red bars show performance degradation from pooling. Most retina-specific features are suppressed.*

Worst cases: *Vein tortuosity (-56%), Artery tortuosity (-48%), Vein fractal dimension (-30%). These features are invisible in DEXA, so pooling destroys this information.*

Conclusions

The hypothesis is confirmed.

JEPA's multi-modal contrastive training trades modality-specific information for cross-modal alignment. This is **not a bug**—it's an inherent property of the L2 minimization objective.

Recommendations for users:

- Use **DEXA-only** for body composition tasks
- Use **Retina-only** for vascular/ocular tasks
- Use **Pooled** for age and general health prediction

For future development:

- Implement disentangled representations (shared vs specific subspaces)
- Add modality-specific auxiliary tasks during training
- Explore attention-based fusion mechanisms

Methods

Data: HPP 10K cohort, ages 40-70

- 8,652 subjects with DEXA scans
- 9,677 subjects with retina scans
- 6,445 subjects with both modalities

Embeddings:

- DEXA-only: 768 dimensions (2 global views concatenated)
- Retina-only: 768 dimensions (2 global views concatenated)
- Pooled: 384 dimensions (mean of 4 global views)

Analysis:

- 5-fold cross-validation, subject-level splits
- Ridge regression (α=1.0)
- Performance metric: **Pearson correlation**

Outcomes tested: 44 total

- Shared: Age, BMI (2)
- DEXA-specific: Body composition (14)
- Retina-specific: Vessel morphology (17)
- Cross-modal: Metabolic, immune, liver, renal (11)